

Melanoma Clustering Explanation

Interpretation note: this model predicts from tabular attributes linked to image ID, not from raw pixels.



Prediction Summary

Image ID: IMD429
Predicted class: Melanoma
Assigned cluster: 1
Cluster mapped label: Melanoma
Cluster melanoma rate: 92.9% (26/28)
Distance to centroid: 2.8182
Latent position (PC1, PC2): 1.630, 1.969
PCA variance (PC1, PC2): 51.7%, 16.6%
Ground truth (if available): Melanoma

Top feature differences vs assigned cluster prototype

Feature	Image value	Cluster prototype	Status
Asymmetry (0/1/2)	0	1.857	different
Dots/Globules (A/AT/T)	A	AT	different
Regression Areas (A/P)	A	P	different
White	NA	X	different
Red	NA	X	different

Cluster-level predicted melanoma fraction: 100.0%. Cluster-to-label mapping: {0: 0, 1: 1, 2: 0}.

Latent Space Position of Test Case

PCA projection of model input space. Star marker is the test case.

